

RITHVIK RAO

Daman & Diu, India

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Rithvik Rao

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Education

Indian Institute of Information Technology, Sri City

Aug. 2023 – May 2027

B.Tech in Computer Science and Engineering

CGPA: 8.90 / 10

Relevant Coursework

- Data Structures
- Database Management
- Web Technologies
- Cloud Computing
- Operating Systems
- HPC
- OOPS
- Algorithm Analysis
- Computer Networks
- Computer Architecture

Experience

SDE Intern

October 2025 – January 2026

Soundverse AI

Remote

- Built and optimized web application using **React.js**, improving performance, responsiveness, and accessibility across devices for an early-stage product serving **100K+ active users**.
- Collaborated closely with backend engineers to integrate and deploy **AI-powered tools and APIs**, strengthening cross-team collaboration and accelerating feature delivery.
- Identified and resolved a large volume of frontend and integration issues, significantly improving application stability, UX consistency, and reducing user-reported bugs across core flows.
- Redesigned the **user cancellation flow** and implemented/debugged **Stripe-based payment logic**, resulting in a **significant reduction in user churn** and improved payment reliability in a high-growth startup environment.

Projects

Pinak

- Built a responsive web and mobile storefront for a cosmetics brand, enabling product discovery, browsing, and a smoother shopping experience for customers.
- Developed key app flows and screens as part of a 3-member team, contributing primarily to the mobile experience while aligning with client requirements and usability goals.
- Collaborated closely with the backend and project team to deliver a functional, user-friendly interface for a real client project under faculty guidance.

MQTT Security Hardening

- Extended the Mochi MQTT broker with TLS session resumption, adaptive TLS profiles, and defense modules against malformed user properties and AUTH floods.
- Created load and attack generator modules to benchmark broker behavior under high concurrency and time-constrained attack scenarios.
- Reduced average reconnect latency from 5.49 ms to 1.80 ms, a 67% improvement, while confirming TLS session reuse across reconnect trials.
- Under malicious property-injection load, reduced peak broker memory from 2,602.95 MB to 22.26 MB (99% reduction)

Replix

- Designed a full-stack WhatsApp automation platform using WhatsApp Web.js, REST APIs, QR-based authentication, and Socket.io for real-time messaging.
- Implemented LLM-powered auto-replies with configurable rules, logging, and analytics through a React + TailwindCSS dashboard.
- Deployed backend services on AWS and the frontend on Vercel, enabling continuous deployment and production availability.

Technical Skills

Languages: Java, JavaScript, TypeScript, SQL

Developer technologies: React, Node.js, Express.js, HTML, Prisma, Next.js

Developer Tools / Cloud: VS Code, Git, Android Studio

Database & Backend: MongoDB, MySQL

Soft Skills: Teamwork, Problem Solving, Communication, Time Management